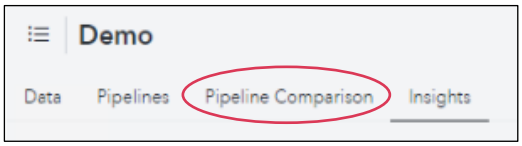
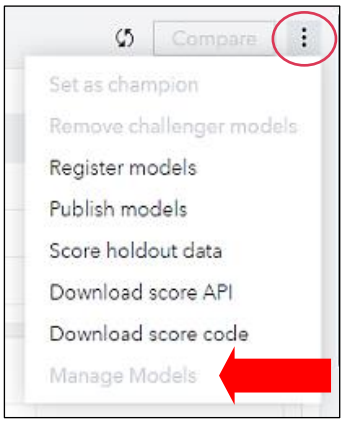
Comparing Multiple Models across Pipelines and Registering the Champion Model

The Pipeline Comparison tab enables you to register the champion model and use it in the Manage Models tool.

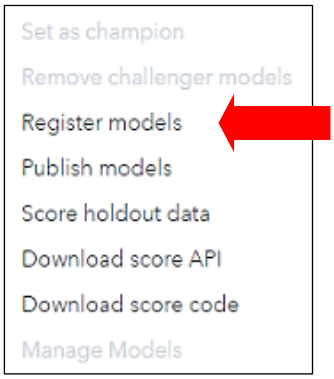
1. Now that we have gained additional insights into the project and the models that it contains, we prepare for model deployment. Return to the pipeline comparison results by clicking the **Pipeline Comparison** tab.



2. Make sure that only the champion model is selected. Click the three vertical dots in the right top corner to access the Project pipeline menu. Note that Manage Models is not available.



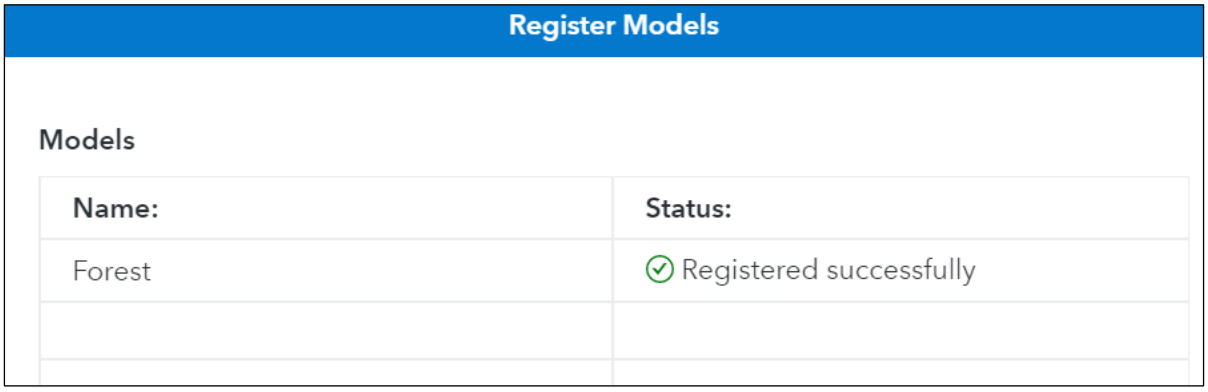
3. Select **Register models**.



The Register Models window appears.

The spinning circle next to **Registering** in the Status column indicates that the selected model (in this case, **Forest**) is actively being registered so that the model is available to other SAS applications.

After registration is complete, the Register Models window is updated to indicate that the registration process has successfully completed.



When the champion model is registered, it can be viewed and used in SAS Model Manager, where you can export the score code in different formats, deploy the model, and manage its performance over time.

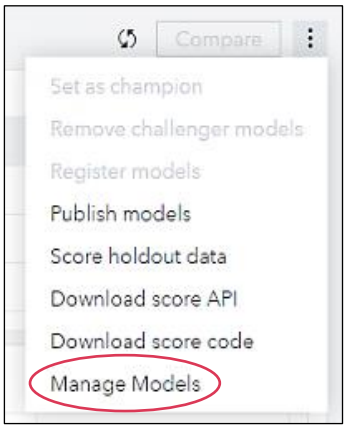
4. Close the Register Model window after the model is registered successfully.

End of Demonstration

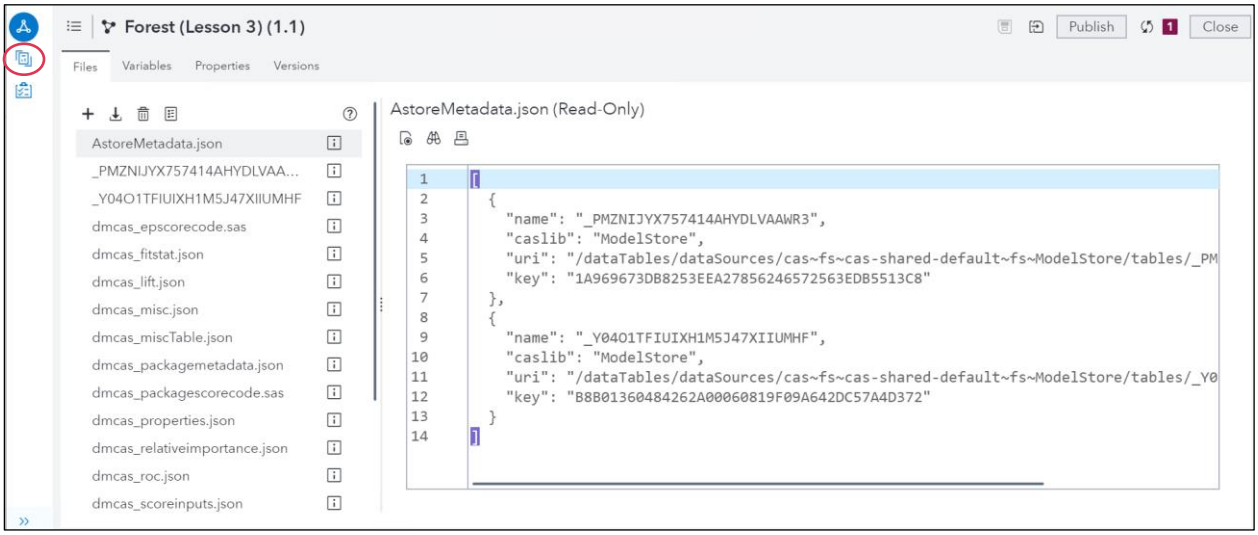
Exploring the Features for Scoring and Running a Scoring Test in Model Manager

In this demonstration, you place the champion model selected by Pipeline Comparison in Model Studio into the SAS Model Manager tool in SAS Viya. The model will be deployed on a scoring data set.

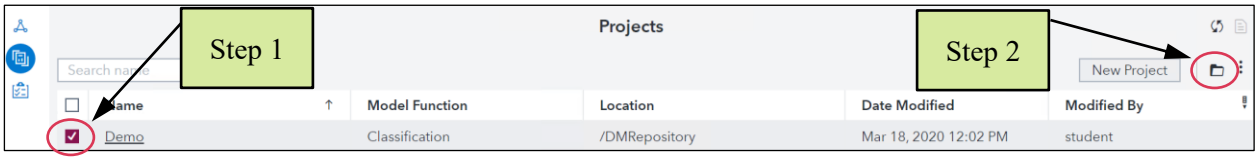
1. In Pipeline Comparison, click the **Project pipeline** menu (three vertical dots). Notice that the Manage Models option is now available because the champion model has been registered. Select **Manage Models**.



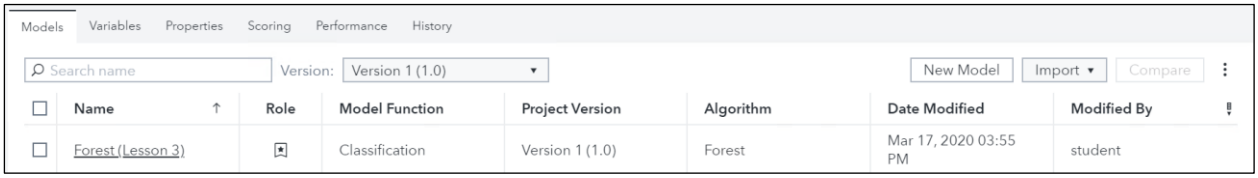
You are redirected to the Manage Model tool. A window containing a list of files is shown, which includes codes for training and scoring the model. Click the second icon assigned to **Projects** in the left pane of the window. This icon takes you to the SAS Model Manager projects.



The SAS Model Manager project named Demo is based on the Model Studio project of the same name. Also in this project are models that were registered within Model Studio. Select the **Demo** project (step 1) and click the **Open** shortcut button in the upper right corner (step 2), or click the project name.



2. Models within the Demo project that have been registered are shown.



There is currently a single model in the Demo project, a forest model. The name of the model, for example, **Forest (Lesson 3)**, is based on the champion model and pipeline name that the model came from in Model Studio.

Across the top of the page is a series of tabs. These tabs are used throughout the model management process, which goes beyond model deployment.

The *Models tab* shows registered models within the Model Manager project.

The *Variables tab* is where input and output variables can be added to project and model objects.

The *Properties tab* contains the project metadata. Project metadata includes information such as the name of the project, the type of project, the project owner, the project location, and which tables and variables are used by project processes, such as scoring. Project properties are organised into General, Tags, and User Defined.

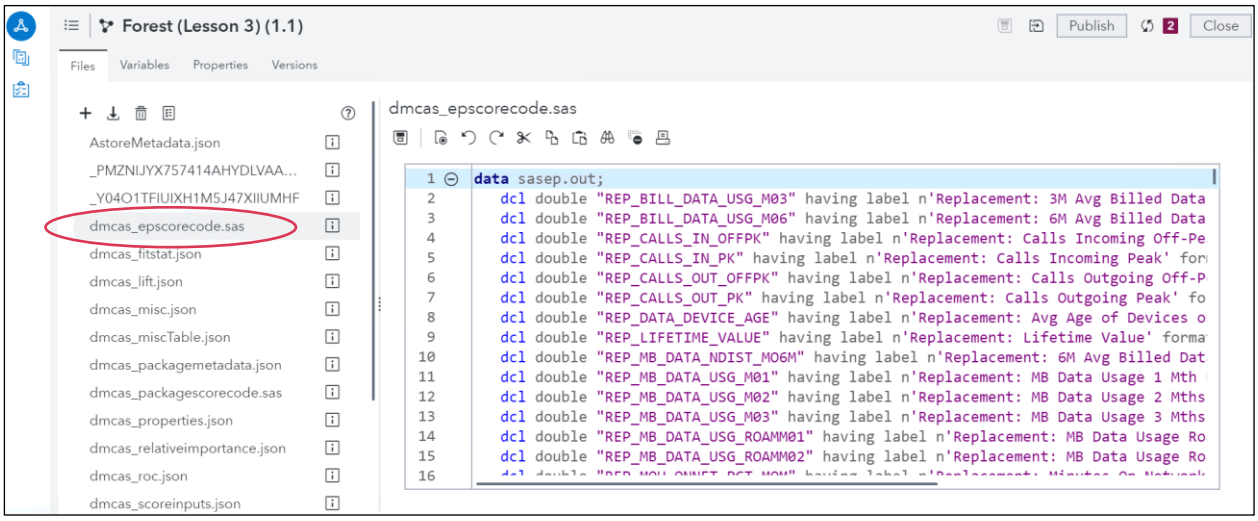
The *Scoring tab* is where scoring tests can be run and also where published models can be validated.

The *Performance tab* shows performance monitoring reports, which are generated from scored data.

The *History tab* shows the history of how the project and model have been used, including information about when they were created, when champion models were defined, and when the model was last deployed.

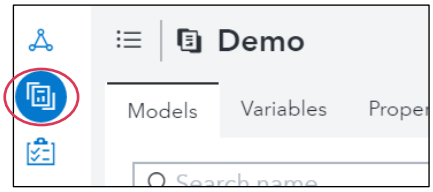
3. Click the name of the model. (Do not click the selection check box in front of it.)

4. Select **dmcas_epscorecode.sas** in the column on the left.

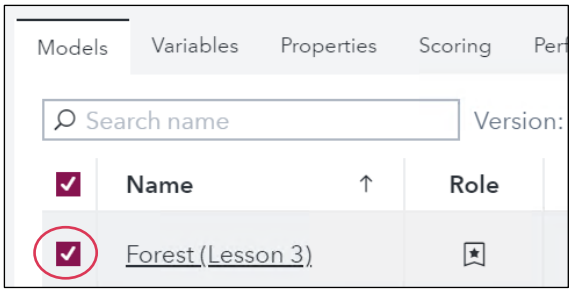


This file is the score code generated by the model and the other data preparation nodes contained within the pipeline. The score code can be exported to deploy the model in production considering distinct environments and platforms.

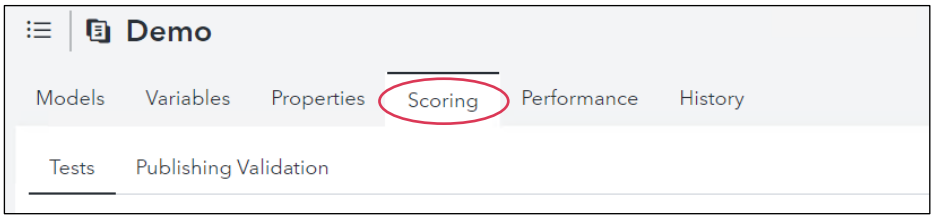
5. Click the **Close** button in the upper right corner of the score code window.
6. Click the second icon for **Projects** on the left pane of the window to return back to the projects view.



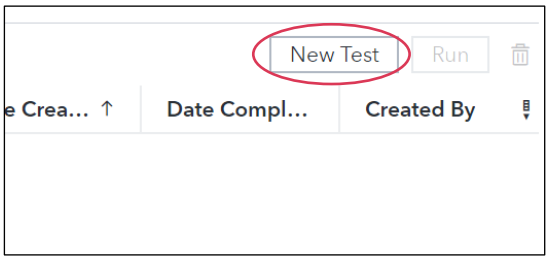
7. If the check box next to the name of the model is not already selected, click it.



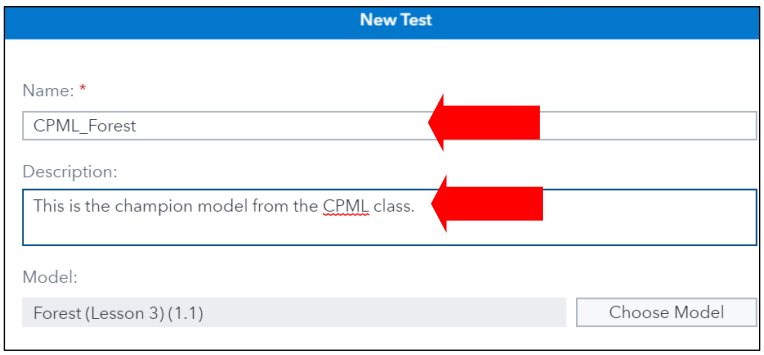
8. Click the **Scoring** tab.



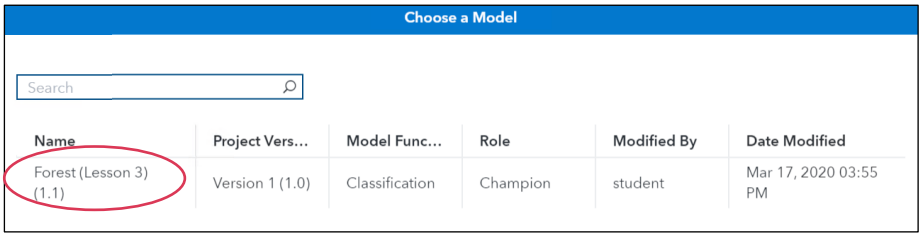
9. In the upper right corner, click **New Test** to create and run a scoring test on the selected model.



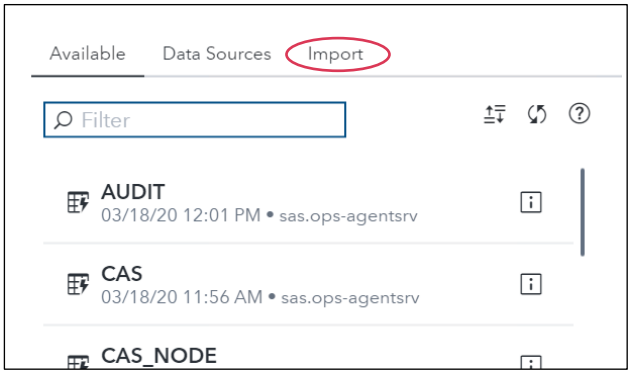
10. Enter **CPML_Forest** as the name and **This is the champion model from the CPML class.** as the description in the New Test window.



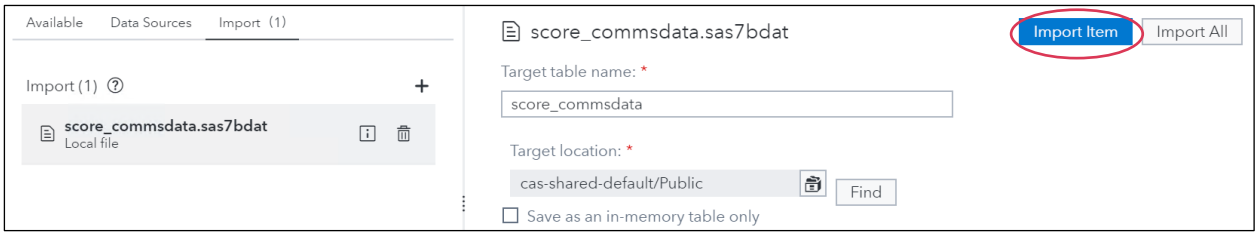
11. Select **Choose Model** in the **Model** field and select the **Forest (Lesson3)** model from the Choose a Model window.



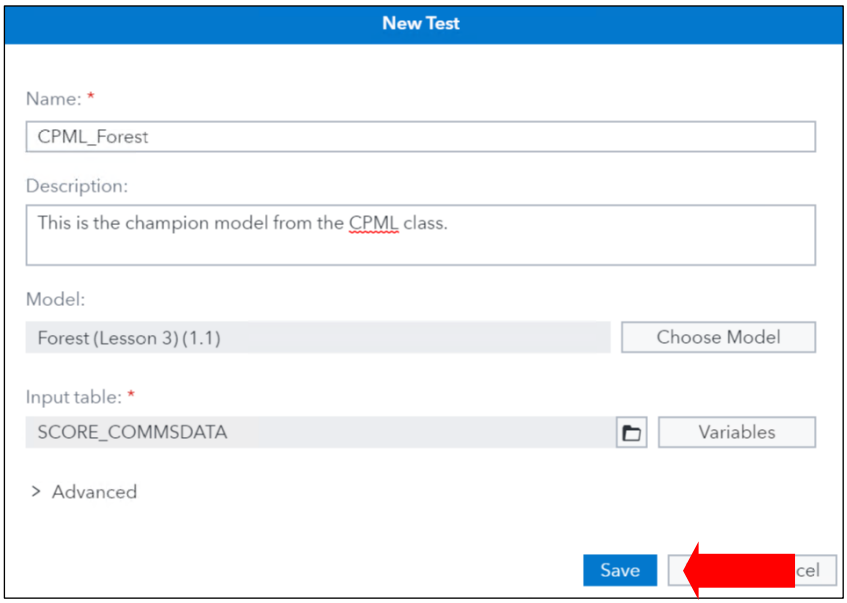
12. Click  (**Select a table**) in the **Input table** field.
13. Click the **Import** tab.



14. Select **Local Files** and then **Local File**. Browse to **D:\Workshop\Winsas\CPML**.
15. Select **score_commsdata.sas7bdat** and then click **Open**.
16. Click **Import Item**.



17. When the table is successfully imported, click **OK**.
18. The New Test window should appear as follows:

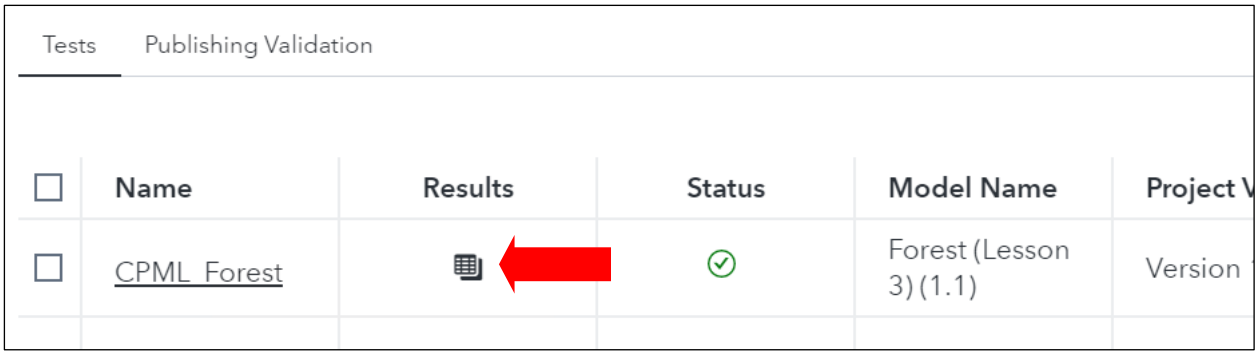


19. Click **Save**.

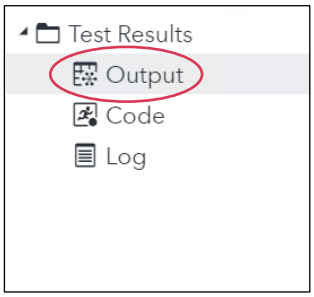
20. First, select the check box next to **CPML_Forest**. Then click **Run** in the upper right corner of the window.



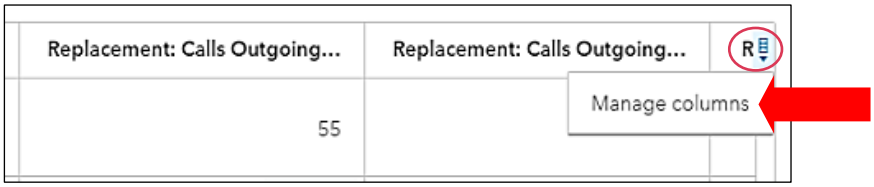
21. Observe that the Status column has a green check mark and that the table icon appears under the Results column. This indicates that the test ran successfully. Click  (**View results**) in the Results column.



22. Click **Output** under Test Results in the left pane to see the predicted scores.



23. Click the **Options** button and then select **Manage columns**.



24. Move all of the variables listed under **Displayed columns** to **Hidden columns** by clicking the **Remove All** shortcut button in the middle of the window.

25. Select the following variables associated with scored output and the Primary Key (which is the Customer ID variable). Click the **Add** shortcut button in the middle of the window to move them from **Hidden columns** to **Displayed columns**. (Hold down the Ctrl key to select multiple variables simultaneously.)

    Into: churn

    Predicted: churn = 0

    Predicted: churn = 1

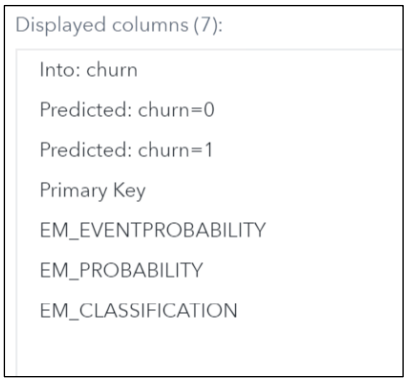
    Primary Key

    EM_EVENTPROBABILITY

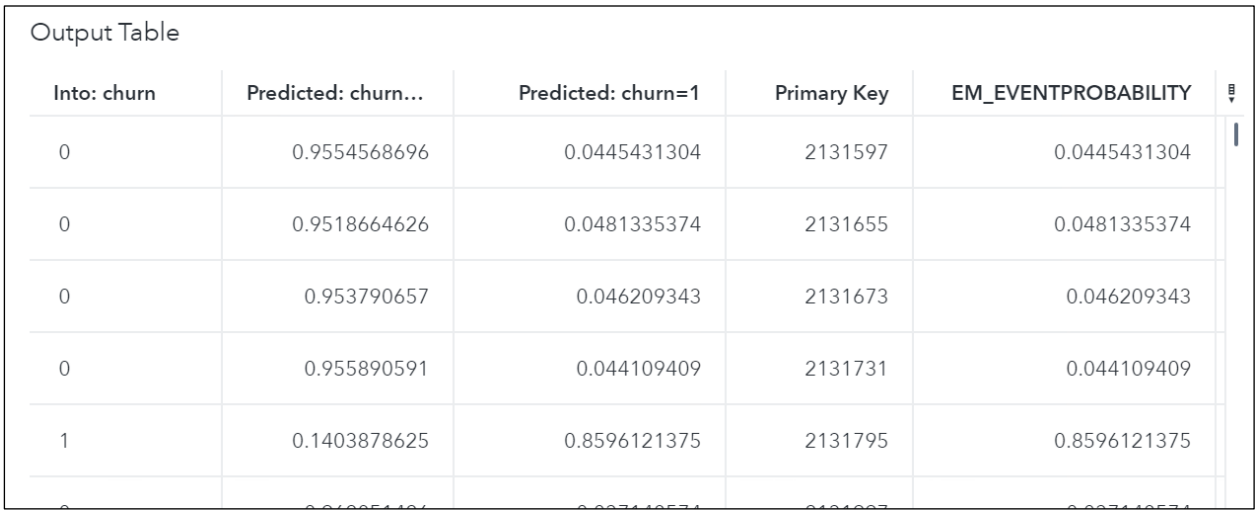
    EM_PROBABILITY

    EM_CLASSIFICATION

Displayed columns should appear as follows:



26. Click **OK** and observe the Output table.



By default, this table is created in the Public library and can be consumed for business use.

27. Close the Output Table window.

End of Demonstration